HOSPITAL MANAGEMENT APPLICATION

Project Description

- This Application allows the patient and doctors to schedule an appointments by using their login credentials.
- This application also allows to store the data and view the prescription.
- In the application limited access is given to doctors and patient according to their appointment related activities.

Tools Used:

- IntelliJ IDEA
- Spring Boot
- Apache Maven
- Docker
- MongoDB
- GitHub

IntelliJ IDEA

- IntelliJ IDEA is a cross-platform IDE that provides consistent experience on the Windows, macOS, and Linux operating systems.
- IntelliJ IDEA is an integrated development environment (IDE) written in Java for developing computer software written in Java, Kotlin, Groovy, and other JVM-based languages.
- It is developed by JetBrains (formerly known as IntelliJ) and is available as an Apache 2 Licensed community edition, and in a proprietary commercial edition. Both can be used for commercial development.

Spring Boot

- Java Spring Framework (Spring Framework) is a popular, open source, enterprise-level framework for creating standalone, production-grade applications that run on the Java Virtual Machine (JVM).
- Java Spring Boot (Spring Boot) is a tool that makes developing web application and microservices with Spring Framework faster and easier through three core capabilities:

1. Autoconfiguration
2. An opinionated approach to configuration
3. The ability to create standalone applications

Apache Maven

- Apache Maven is a software project management and comprehension tool.
- Based on the concept of a project object model (POM). Maven can manage a project's build, reporting and documentation from a central piece of information.
- The Maven project is hosted by the Apache Software Foundation, where it was formerly part of the Jakarta Project.

Docker

- Docker is an open platform for developing, shipping, and running applications.
- Docker enables you to separate your applications from your infrastructure so you can deliver software quickly.
- With Docker, you can manage your infrastructure in the same ways you manage your applications.
- By taking advantage of Docker’s methodologies for shipping, testing, and deploying code quickly, you can significantly reduce the delay between writing code and running it in production.

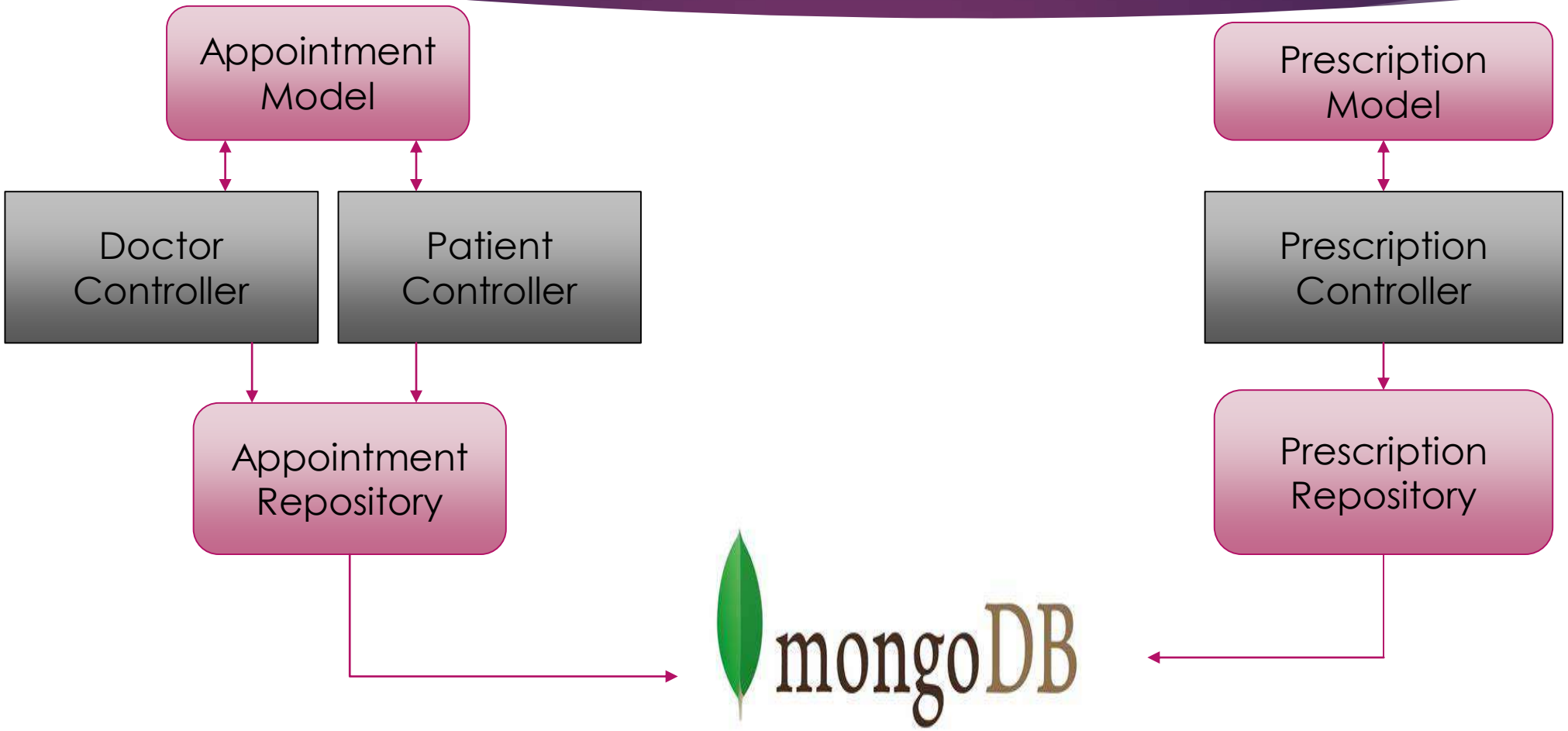
Mongo DB

- **MongoDB** is a source-available cross-platform document-oriented database program. Classified as a NoSQL database program, MongoDB uses JSON-like documents with optional schemas.
- MongoDB is developed by MongoDB Inc. and licensed under the Server Side Public License (SSPL) which is deemed non-free by several distributions. MongoDB is a member of the MACH Alliance.
- MongoDB stores data in flexible, JSON-like documents, meaning fields can vary from document to document and data structure can be changed over time

GitHub

- Git is an open-source, version control tool created in 2005 by developers working on the Linux operating system.
- GitHub is a company founded in 2008 that makes tools which integrate with git. You do not need GitHub to use git, but you cannot use GitHub without using git.
- GitHub is maintained by microsoft.

Application Architecture



Model

- Appointment model: This is the first model of the microservices that stores important data related to appointments such as appointmentId, patientName, doctorName, date as well as the prescription object.
- Prescription model: This is the second model of the microservice that stores important data related to prescriptions such as prescriptionId, patientName, doctorName, appointmentId and its description.

Controllers

- DoctorController
- PatientController
- PrescriptionController

DoctorController

The DoctorController has the following two Rest EndPoints:

- /doctor/doctorappointment: This is a GET method which is used to read the stored data of patient from the database. It uses doctor name as parameter.
- /doctor/savedocappointment: This is a POST method which is used to stored the patient data into the database.

PatientController

The PatientController has the following two Rest EndPoints:

- /patient/myappointment: This is the GET method which is used to read the data when the patient wants to get some data from the database. It uses patient name as a parameter.
- /patient/savepatientappointment: This is a POST method which is used to save the data when patient want to stored some data into the database or schedule some appointment.

PrescriptionController

The PrescriptionController has the following two Rest EndPoints:

- /prescription/viewprescription: This is a GET method which is used to view the prescription data stored in the mongo database.
- /prescription/saveprescription: This is a POST method which is used to stored the prescription details in the mongo database.

Repositories:

- AppointmentRepository: This database is used to stored the appointment data into the mongo database. It extends the mongo repository and consists of various entities that are used to store appointment data.

- PrescriptionRepository: This repository is used to store prescription data into mongo database. This also extends the mongo repository and consist of various entities that are used to stored the prescription data.

Thank you!